

```
File Edit Selection View Go Run Terminal Help
DSA assignment
Question2.java Question1.java Question3.java Question4.java
Week3 > Question1.java > Question1 > main(String[])
1 package Week3;
2 import java.util.*;
3
4 public class Question1 {
5
6     static void mergeSort(int[] arr, int left, int right) {
7         if (left < right) {
8             int mid = left + (right - left) / 2;
9
10            mergeSort(arr, left, mid);
11            mergeSort(arr, mid + 1, right);
12
13            merge(arr, left, mid, right);
14        }
15    }
16
17    static void merge(int[] arr, int left, int mid, int right) {
18
19        int n1 = mid - left + 1;
20        int n2 = right - mid;
21
22        int[] L = new int[n1];
23        int[] R = new int[n2];
24
25        for (int i = 0; i < n1; i++)
26            L[i] = arr[left + i];
27
28        for (int j = 0; j < n2; j++)
29            R[j] = arr[mid + 1 + j];
30
31        int i = 0, j = 0, k = left;
32
33        while (i < n1 && j < n2) {
34            if (L[i] <= R[j])
35                arr[k++] = L[i++];
36            else
37                arr[k++] = R[j++];
38        }
39
40        while (i < n1)
41            arr[k++] = L[i++];
42        while (j < n2)
43            arr[k++] = R[j++];
44    }
45
46    public static void main(String[] args) {
47        int arr[] = {2, 7, 8, 9, 10};
48        mergeSort(arr, 0, arr.length - 1);
49        for (int i : arr)
50            System.out.print(i + " ");
51        System.out.println();
52    }
53}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

PS C:\Users\Vinay singh\DSA assignment\Week3> java Question1
2 7 8 9 10
PS C:\Users\Vinay singh\DSA assignment\Week3>

Ln 64, Col 1

Rainy days ahead
30°C

```
Question2.java Question1.java Question3.java Question4.java
Week3 > Question1.java > Question1 > main(String[])
4 public class Question1 {
17     static void merge(int[] arr, int left, int mid, int right) {
33         while (i < n1 && j < n2) {
34             if (L[i] <= R[j])
35                 arr[k++] = L[i++];
36             else
37                 arr[k++] = R[j++];
38         }
39
40         while (i < n1)
41             arr[k++] = L[i++];
42
43         while (j < n2)
44             arr[k++] = R[j++];
45     }
46
47     Run | Debug
48     public static void main(String[] args) {
49
50         Scanner sc = new Scanner(System.in);
51
52         int n = sc.nextInt();
53
54         int[] arr = new int[n];
55
56         for (int i = 0; i < n; i++)
57             arr[i] = sc.nextInt();
58
59         mergeSort(arr, left: 0, n - 1);
60
61         for (int x : arr)
62             System.out.print(x + " ");
63
64         sc.close();
65     }
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

- PS C:\Users\Winay singh\DSA assignment\
- PS C:\Users\Winay singh\DSA assignment\
- 5
- 8
- 7
- 9
- 19
- 2
- 2 7 8 9 19
- PS C:\Users\Winay singh\DSA assignment\

Java: Ready

2 cm of rain Thursday

Search

```
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DSA assignment

Question2.java Question3.java Question4.java
Week3 > Question2.java > Question2 > main(String[])
4
5 public class Question2 {
6
7     static void mergeSort(int[] arr, int left, int right) {
8         if (left < right) {
9             int mid = left + (right - left) / 2;
10
11             mergeSort(arr, left, mid);
12             mergeSort(arr, mid + 1, right);
13
14             merge(arr, left, mid, right);
15         }
16     }
17
18     static void merge(int[] arr, int left, int mid, int right) {
19
20         int n1 = mid - left + 1;
21         int n2 = right - mid;
22
23         int[] L = new int[n1];
24         int[] R = new int[n2];
25
26         for (int i = 0; i < n1; i++)
27             L[i] = arr[left + i];
28
29         for (int j = 0; j < n2; j++)
30             R[j] = arr[mid + 1 + j];
31
32         int i = 0, j = 0, k = left;
33
34         while (i < n1 && j < n2) {
35             if (L[i] <= R[j])
36                 arr[k++] = L[i++];
37             else
38                 arr[k++] = R[j++];
39         }
40     }
41 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

- PS C:\Users\Vinay singh\DSA assignment\Week3> java Question2
- PS C:\Users\Vinay singh\DSA assignment\Week3> java Question2

4
89
76
45
21
21 45 76 89
Median: 60
Orders Above Median: 2
Difference: 88

PS C:\Users\Vinay singh\DSA assignment\Week3>

2 cm of rain
Thursday


```

34 while (i < n1 && j < n2) {
35     if (L[i] <= R[j])
36         arr[k++] = L[i++];
37     else
38         arr[k++] = R[j++];
39 }
40
41 while (i < n1)
42     arr[k++] = L[i++];
43
44 while (j < n2)
45     arr[k++] = R[j++];
46 }
47

```

Run | Debug

```

48 public static void main(String[] args) {
49
50     Scanner sc = new Scanner(System.in);
51
52     int n = sc.nextInt();
53
54     int[] arr = new int[n];
55
56     for (int i = 0; i < n; i++)
57         arr[i] = sc.nextInt();
58
59     mergeSort(arr, left: 0, n - 1);
60
61     for (int x : arr)
62         System.out.print(x + " ");
63     System.out.println();
64
65     int median;
66     if (n % 2 == 1)
67         median = arr[n / 2];
68     else
69         median = (arr[n / 2 - 1] + arr[n / 2]) / 2;
70
71     int count = 0;
72     for (int x : arr)
73         if (x > median)
74             count++;
75
76     int diff = arr[n - 1] - arr[0];
77
78     System.out.println("Median: " + median);
79     System.out.println("Orders Above Median: " + count);
80     System.out.println("Difference: " + diff);
81
82     sc.close();
83 }
84 }
85

```

```

1 package Week3;
2
3 import java.util.*;
4
5 public class Question3 {
6
7     static int partition(int[] arr, int low, int high) {
8         int pivot = arr[high];
9         int i = low - 1;
10
11         for (int j = low; j < high; j++) {
12             if (arr[j] <= pivot) {
13                 i++;
14                 int temp = arr[i];
15                 arr[i] = arr[j];
16                 arr[j] = temp;
17             }
18         }
19
20         int temp = arr[i + 1];
21         arr[i + 1] = arr[high];
22         arr[high] = temp;
23
24         return i + 1;
25     }
26
27     static void quickSort(int[] arr, int low, int high) {
28         if (low < high) {
29             int pi = partition(arr, low, high);
30
31             quickSort(arr, low, pi - 1);
32             quickSort(arr, pi + 1, high);
33         }
34     }
35
36     public static void main(String[] args) {
37         Scanner sc = new Scanner(System.in);
38
39         int n = sc.nextInt();
40         int[] arr = new int[n];
41
42         for (int i = 0; i < n; i++)
43             arr[i] = sc.nextInt();
44
45         quickSort(arr, low: 0, n - 1);
46
47         for (int x : arr)
48             System.out.print(x + " ");
49         sc.close();
50     }
51 }
52

```

DSA assignment

PROBLEMS	OUTPUT	DEBUG CONSOLE	TERMINAL	POINTS
PS C:\Users\Winey s\gh\DSA assignment\Week3> javac Question3.java				5
PS C:\Users\Winey s\gh\DSA assignment\Week3> java Question3.java				45
				54
				67
				78
				32
				32 45 54 67 78
PS C:\Users\Winey s\gh\DSA assignment\Week3>				

I

```
File Edit Selection View Go Run Terminal Help
Question2.java Question3.java Question4.java X DSA assignment
Week3 > Question4.java > Question4 > main(String[])
1 package Week3;
2
3 import java.util.*;
4
5 public class Question4 {
6
7     static int partition(int[] arr, int low, int high) {
8
9         int pivot = arr[high];
10        int i = low - 1;
11
12        for (int j = low; j < high; j++) {
13            if (arr[j] >= pivot) {
14                i++;
15                int temp = arr[i];
16                arr[i] = arr[j];
17                arr[j] = temp;
18            }
19
20            int temp = arr[i + 1];
21            arr[i + 1] = arr[high];
22            arr[high] = temp;
23
24            return i + 1;
25        }
26
27        static void quickSort(int[] arr, int low, int high) {
28
29            if (low < high) {
30                int pi = partition(arr, low, high);
31
32                quickSort(arr, low, pi - 1);
33                quickSort(arr, pi + 1, high);
34            }
35        }
36
37        Run|Debug
38        public static void main(String[] args) {
39
40            Scanner sc = new Scanner(System.in);
41
42            PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
43            • PS C:\Users\Vinay singh\DSA assignment\Week3> javac Question4.java
44            • PS C:\Users\Vinay singh\DSA assignment\Week3> java Question4.java
45            5
46            43
47            45
48            67
49            87
50            98
51            98 87 67 45 43
52            Top 5: 98 87 67 45 43
53            Average of Top 5: 68
54            Values Above Overall Average: 2
55            • PS C:\Users\Vinay singh\DSA assignment\Week3>
```



```
File Edit Selection View Go Run Terminal Help
Question2.java Question3.java Question4.java x
Week3 > Question4.java > Question4 > main(String[])
5 public class Question4 {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int[] arr = new int[n];
        long sum = 0;
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
            sum += arr[i];
        }
        quickSort(arr, low: 0, n - 1);
        for (int x : arr)
            System.out.print(x + " ");
        System.out.println();
        System.out.print(sc: "Top 5: ");
        long topSum = 0;
        for (int i = 0; i < 5; i++) {
            System.out.print(arr[i] + " ");
            topSum += arr[i];
        }
        System.out.println();
        System.out.println("Average of Top 5: " + (topSum / 5));
        double overallAvg = (double) sum / n;
        int count = 0;
        for (int x : arr)
            if (x > overallAvg)
                count++;
        System.out.println("Values Above Overall Average: " + count);
        sc.close();
    }
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

- PS C:\Users\Vinay singh\DSA assignment\Week3> javac Question4.java
- PS C:\Users\Vinay singh\DSA assignment\Week3> java Question4.java

5
43
45
67
87
98
98 87 67 45 43
Top 5: 98 87 67 45 43
Average of Top 5: 68
Values Above Overall Average: 2
PS C:\Users\Vinay singh\DSA assignment\Week3>

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Question2.java Question3.java Question4.java Question5.java

DSA assignment

Week3 > Question5.java > Question5 > mainString@

```
1 package Week3;
2
3 import java.util.*;
4
5 public class Question5 {
6
7     static void heapify(int arr[], int n, int i) {
8
9         int largest = i;
10        int left = 2 * i + 1;
11        int right = 2 * i + 2;
12
13        if (left < n && arr[left] > arr[largest])
14            largest = left;
15
16        if (right < n && arr[right] > arr[largest])
17            largest = right;
18
19        if (largest != i) {
20
21            int temp = arr[i];
22            arr[i] = arr[largest];
23            arr[largest] = temp;
24
25            heapify(arr, n, largest);
26        }
27    }
28
29    static void heapSort(int arr[]) {
30
31        int n = arr.length;
32
33        // Build Max Heap
34        for (int i = n / 2 - 1; i >= 0; i--)
35            heapify(arr, n, i);
36
37        // Heap Sort
38        for (int i = n - 1; i > 0; i--) {
39
40            int temp = arr[0];
41            arr[0] = arr[i];
42            arr[i] = temp;
43
44            heapify(arr, i, 0);
45        }
46    }
47 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\Vinay singh\DSA assignment\Week3> javac Question5.java
5
211
23
45
65
43
23 43 45 65 211
PS C:\Users\Vinay singh\DSA assignment\Week3>
```

Java IDE

Finance headline
India reported 6

Search

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DSA assignment

Question2.java Question3.java Question4.java Question5.java X

Week3 > Question5.java > Question5 > main(String[])

```
5 public class Question5 {
29     static void heapSort(int arr[]) {
30
31         int n = arr.length;
32
33         // Build Max Heap
34         for (int i = n / 2 - 1; i >= 0; i--)
35             heapify(arr, n, i);
36
37         // Heap Sort
38         for (int i = n - 1; i > 0; i--) {
39
40             int temp = arr[0];
41             arr[0] = arr[i];
42             arr[i] = temp;
43
44             heapify(arr, i, 0);
45         }
46     }
47
48     Run|Debug
49     public static void main(String[] args) {
50
51         Scanner sc = new Scanner(System.in);
52
53         int n = sc.nextInt();
54
55         int arr[] = new int[n];
56
57         for (int i = 0; i < n; i++)
58             arr[i] = sc.nextInt();
59
60         heapSort(arr);
61
62         for (int x : arr)
63             System.out.print(x + " ");
64         sc.close();
65     }
66 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

- PS C:\Users\Vinay singh\DSA assignment\Week3> javac Question5.java
- PS C:\Users\Vinay singh\DSA assignment\Week3> java Question5.java

5
211
23
45
65
43
23 43 45 65 211

PS C:\Users\Vinay singh\DSA assignment\Week3>

```
File Edit Selection View Go Run Terminal Help
DSA assignment
Question2.java Question3.java Question4.java Question5.java
Week3 > Question6.java > main(String[])
4
5 public class Question6 {
6
7     static void heapify(int arr[], int n, int i) {
8
9         int largest = i;
10        int left = 2 * i + 1;
11        int right = 2 * i + 2;
12
13        if (left < n && arr[left] > arr[largest])
14            largest = left;
15
16        if (right < n && arr[right] > arr[largest])
17            largest = right;
18
19        if (largest != i) {
20
21            int temp = arr[i];
22            arr[i] = arr[largest];
23            arr[largest] = temp;
24
25            heapify(arr, n, largest);
26        }
27    }
28
29    static void heapSort(int arr[]) {
30
31        int n = arr.length;
32
33        // Build Heap
34        for (int i = n / 2 - 1; i >= 0; i--)
35            heapify(arr, n, i);
36
37        // Sort
38        for (int i = n - 1; i > 0; i--) {
39
40            int temp = arr[0];
41            arr[0] = arr[i];
42            arr[i] = temp;
43
44            heapify(arr, i, 0);
45        }
46    }
47
48    Run | Debug
49    java C:\Users\Vinay singh\DSA assignment\Week3> java Question6.java
50    6
51    54
52    7
53    98
54    99
55    34
56    78
57    7 34 54 78 98 99
58    Fastest: 7
59    Slowest: 99
60    Average: 61.67
61    Cases Faster Than Average: 3
62    Percentage: 50.00%
63    PS C:\Users\Vinay singh\DSA assignment\Week3>
```

```
File Edit Selection View Go Run Terminal Help
Question2.java Question3.java Question4.java Question6.java X
Week3 > Question6.java > Question6 > main(String[])
5 public class Question6 {
48     public static void main(String[] args) {
49
50         Scanner sc = new Scanner(System.in);
51
52         int n = sc.nextInt();
53
54         int arr[] = new int[n];
55
56         int sum = 0;
57
58         for (int i = 0; i < n; i++) {
59             arr[i] = sc.nextInt();
60             sum += arr[i];
61         }
62
63         heapSort(arr);
64
65         // Sorted Array
66         for (int x : arr)
67             System.out.print(x + " ");
68         System.out.println();
69
70         int fastest = arr[0];
71         int slowest = arr[n - 1];
72
73         double avg = (double) sum / n;
74
75         int count = 0;
76
77         for (int x : arr)
78             if (x < avg)
79                 count++;
80
81         double percentage = (count * 100.0) / n;
82
83         System.out.println("Fastest: " + fastest);
84         System.out.println("Slowest: " + slowest);
85         System.out.printf("Average: %.2f\n", avg);
86         System.out.println("Cases Faster Than Average: " + count);
87         System.out.printf("Percentage: %.2f%%\n", percentage);
88         sc.close();
89     }
90 }
91
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

- PS C:\Users\Viney singh\DSA assignment\Week3> javac Question6.java
- PS C:\Users\Viney singh\DSA assignment\Week3> java Question6.java

6
54
7
98
99
34
78
7 34 54 78 98 99
Fastest: 7
Slowest: 99
Average: 61.67
Cases Faster Than Average: 3
Percentage: 50.00%

PS C:\Users\Viney singh\DSA assignment\Week3>

30°C
Partly sunny

Search